

Medical Diagnostic Retrieval System

Milestone 1 — Schema Design & DDL

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Track: A — RAG Pipeline (Retrieval-Augmented Generation)

Database Engine: PostgreSQL 16 — **Vector Store:** pgvector

GitHub: <https://github.com/Arunyaiitm/medical-diagnostic-retrieval-system>

1. Project Description

A RAG-powered system that lets doctors query patient diagnostic records using natural language instead of SQL. Data comes from the NIH Chest X-Ray Dataset (Kaggle, 5,606 records with patient demographics and 14 disease labels), with supporting relational data generated via a documented seed script. The system combines structured SQL filtering with semantic similarity search over diagnostic notes using pgvector.

2. Database Schema (7 Tables, 3NF)

Table	Primary Key	Foreign Keys	Data Source
departments	department_id	—	Seed script
doctors	doctor_id	department_id → departments	Seed script
patients	patient_id	—	NIH dataset
visits	visit_id	patient_id → patients, doctor_id → doctors	NIH + seed
scans	scan_id	visit_id → visits	NIH dataset
diagnoses	diagnosis_id	scan_id → scans, doctor_id → doctors	NIH dataset
prescriptions	prescription_id	diagnosis_id → diagnoses	Seed script

3. Relationships

Entity A	Cardinality	Relationship	Entity B	A Side	B Side
Departments	1 : M	has	Doctors	Partial	Total
Patients	1 : M	attends	Visits	Partial	Total
Doctors	1 : M	conducts	Visits	Partial	Total
Visits	1 : M	produces	Scans	Partial	Total
Scans	1 : M	yields	Diagnoses	Partial	Total
Doctors	1 : M	diagnosed by	Diagnoses	Partial	Total
Diagnoses	1 : M	treated by	Prescriptions	Partial	Total

4. Constraints

- 7 Primary Keys, 7 Foreign Keys (with ON DELETE CASCADE / RESTRICT)
- CHECK constraints: gender (M/F), age (0–130), severity (MILD/MODERATE/SEVERE/CRITICAL), visit_type, view_position, finding_label (14 NIH diseases + No Finding), duration_days (1–365)
- UNIQUE constraints: department_name, email, image_filename
- 10 B-Tree indexes on frequently queried and JOIN columns

5. Data Source

Primary: NIH Chest X-Ray Dataset (Sample) from Kaggle — 5,606 de-identified records with patient ID, age, gender, 14 disease labels, follow-up numbers, view position, and image metadata.

Supporting: Documented Python seed script generating doctors, departments, visits, and prescriptions with realistic medical constraints (specialisation-to-diagnosis mapping, chronological visit ordering, real medication-to-condition pairing).

6. Deliverables

- `schema.sql` — runs from scratch on PostgreSQL 16 with zero errors (tested)
- `er_diagram.pdf` — Chen notation ER diagram with all entities, attributes (PK underlined, FK italic), relationship diamonds, 1:M cardinality, partial/total participation
- Page 2 of ER PDF contains 3NF normalisation argument with functional dependencies + relationship summary table